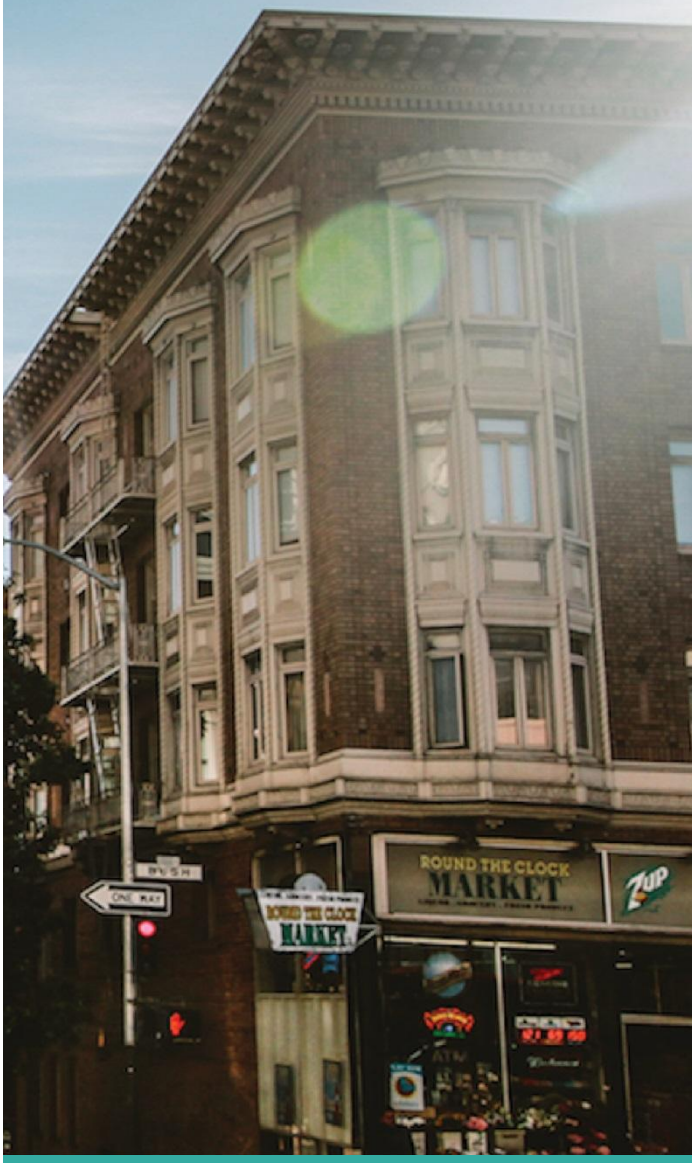
# Assignment 1
# Lab 3&4

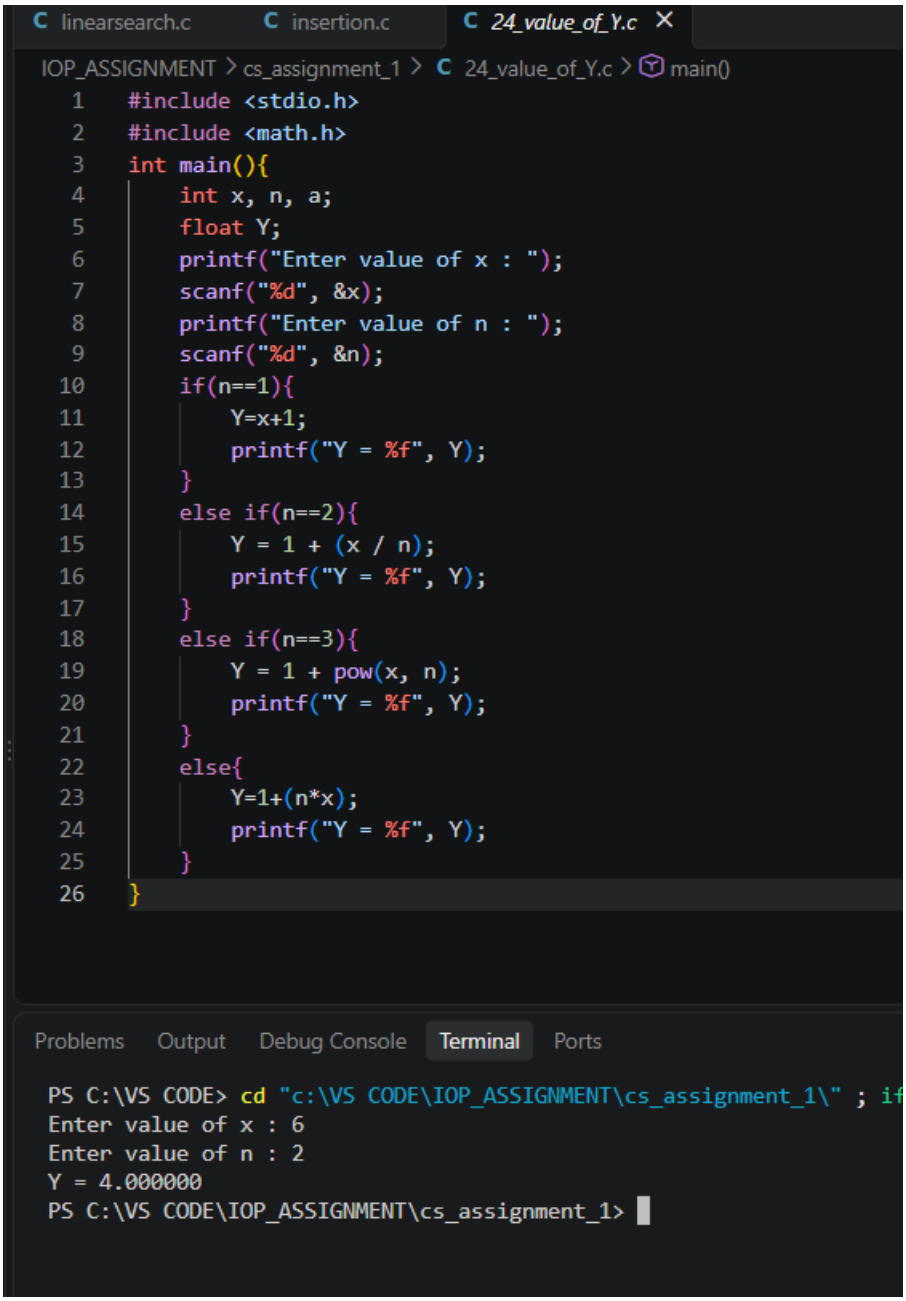



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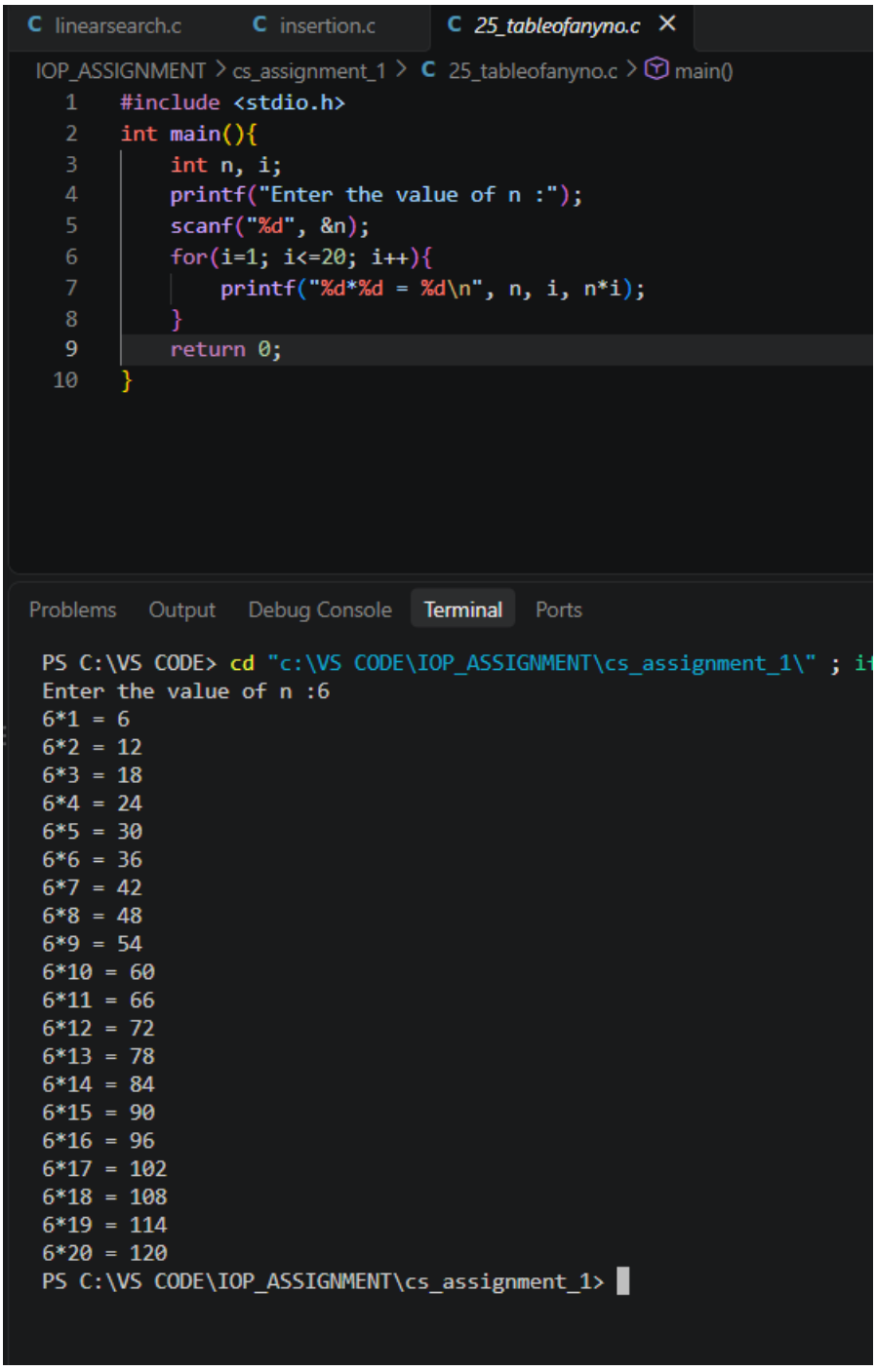
**NIHIT BHAGWAT**
**AI BTECH Div-B**

## 24. WRITE A C PROGRAM TO FIND THE VALUE OF Y USING

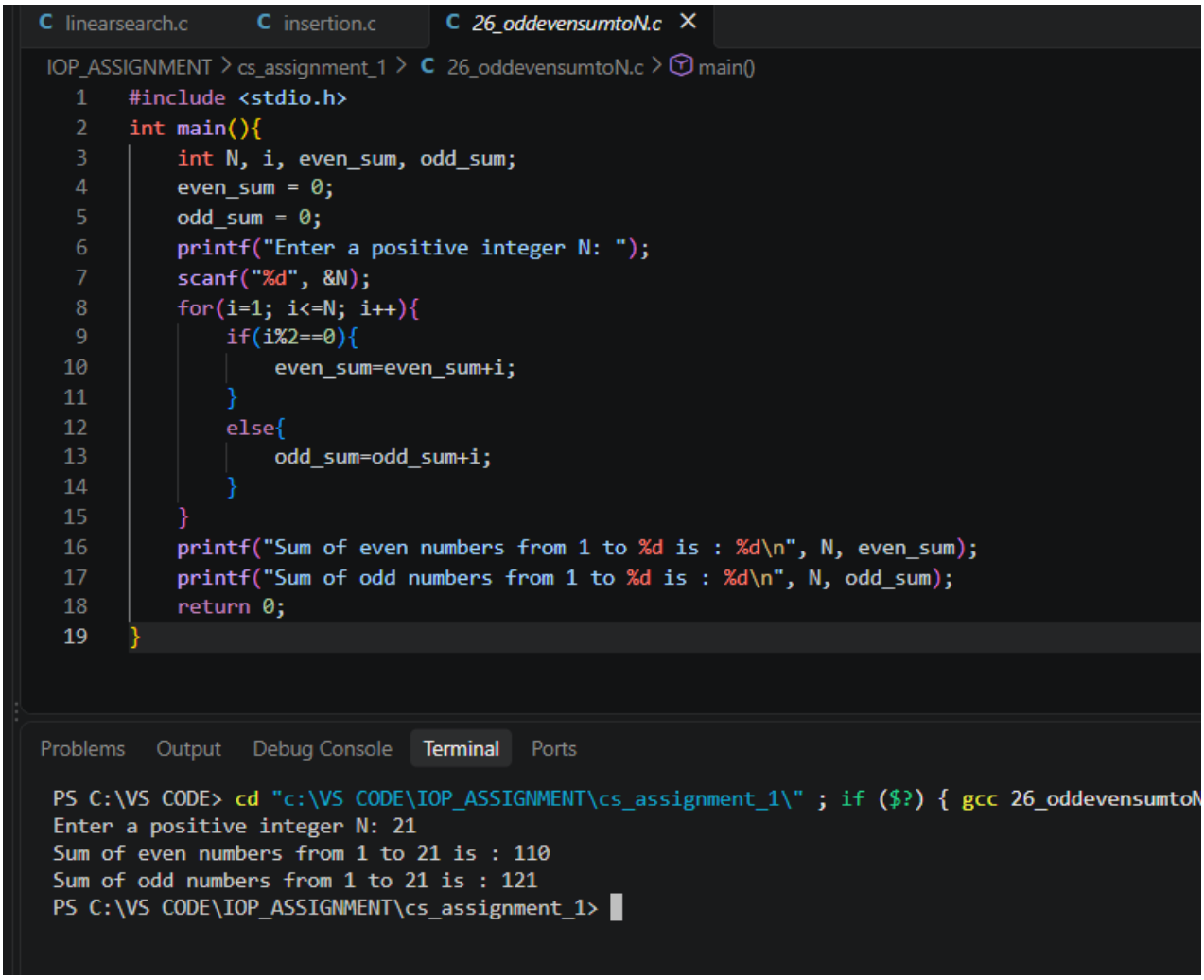
$$Y(x,n) = \begin{cases} 1+x & \text{where } n=1 \\ 1+x/n & \text{where } n=2 \\ 1+x^n & \text{when } n=3 \\ 1+nx & \text{when } n>3 \text{ or } n<1 \end{cases}$$



## 25. PROGRAM TO PRINT TABLE OF ANY NO.



## 26. PROGRAM TO PRINT SUM OF ALL ODD AND EVEN NO B/W 1 TO N.



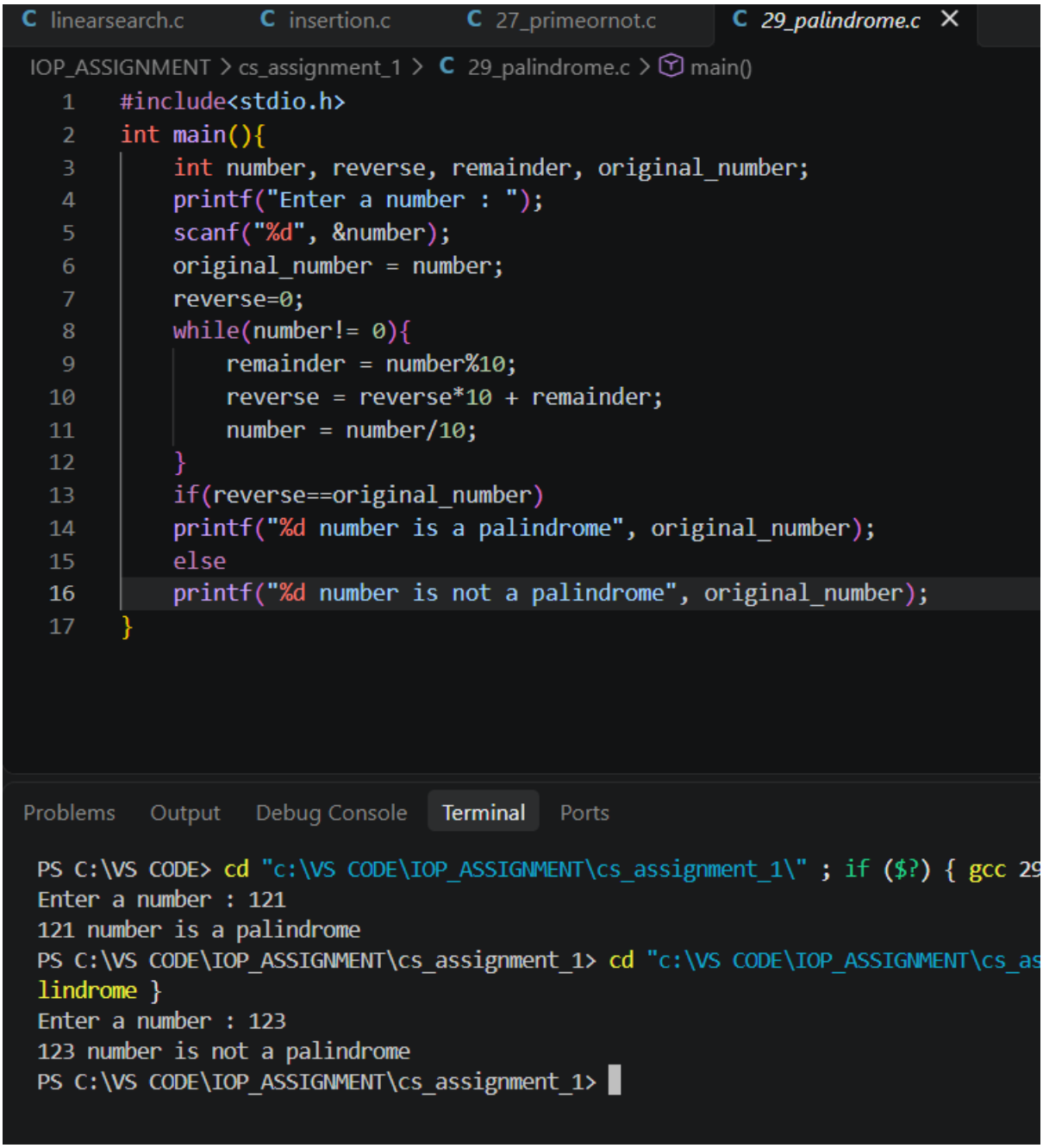
## 27. PROGRAM TO CHECK WHETHER A NO IS PRIME OR NOT.



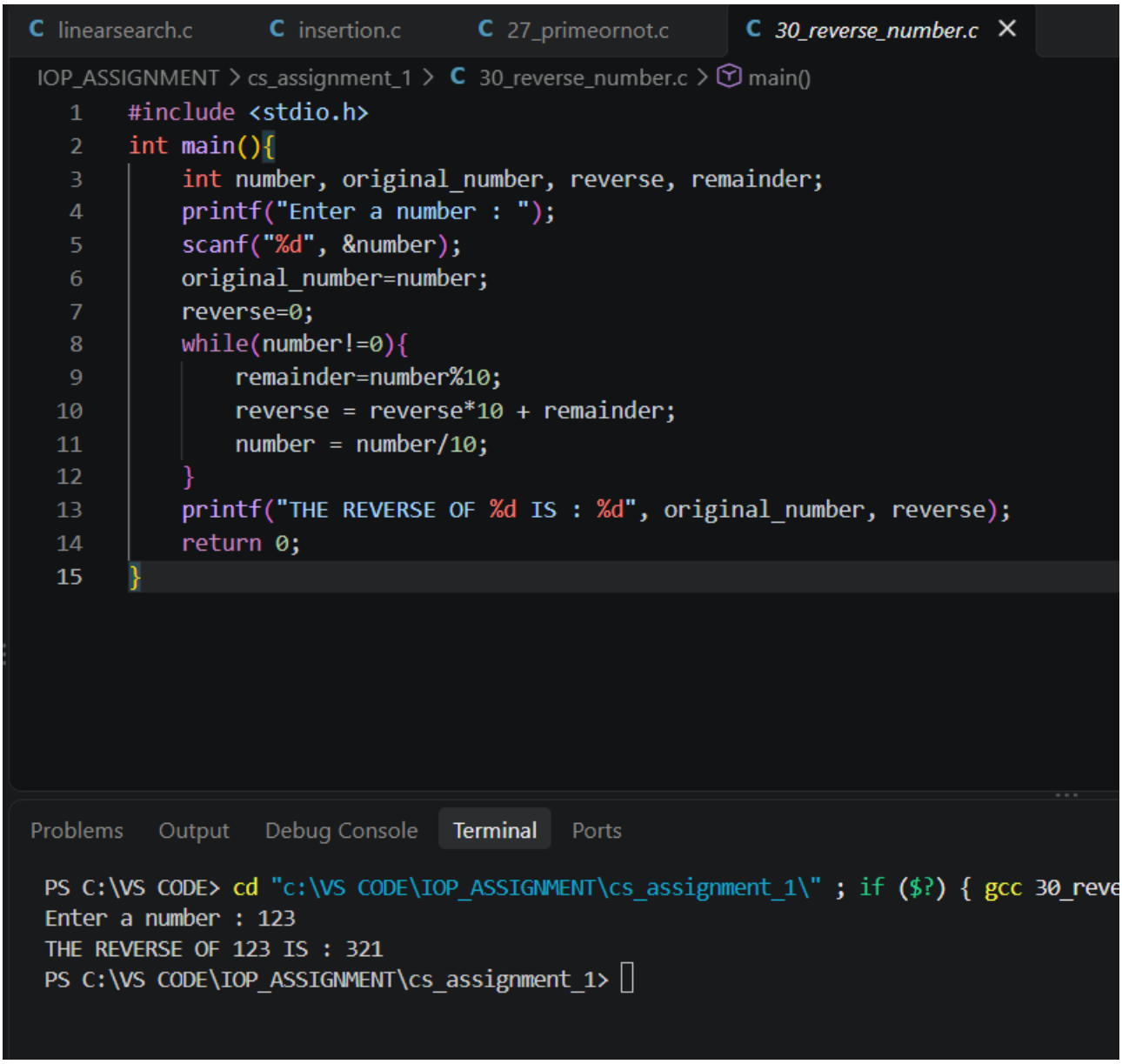
## 28. PROGRAM TO CHECK WHETHER A NO IS ARMSTRONG OR NOT.



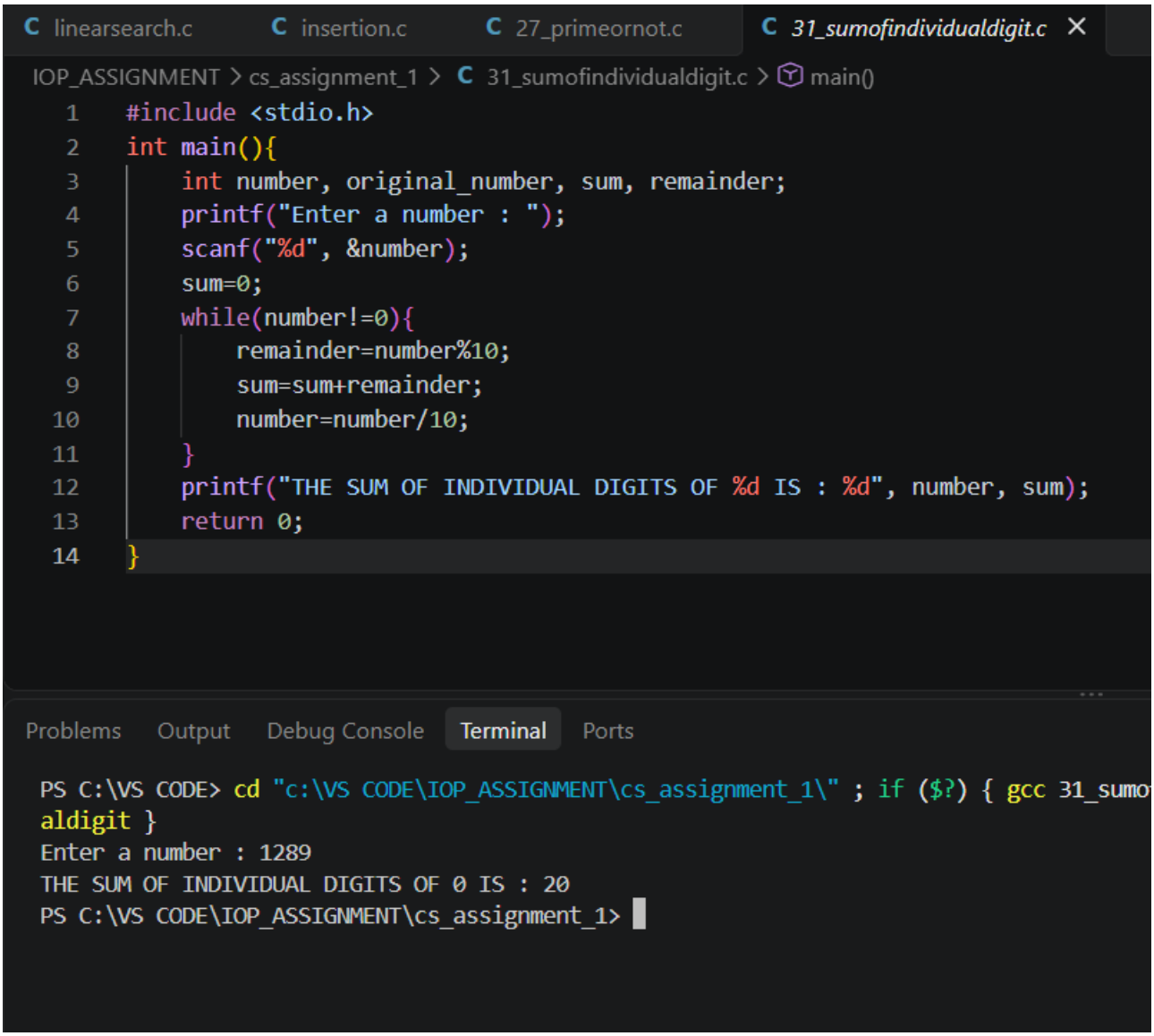
## 29. PROGRAM TO CHECK WHETHER A NO IS PALINDROM OR NOT.



## 30. PROGRAM TO REVERSE OF A GIVEN NO.



## 31. PROGRAM TO PRINT SUM OF INDIVIDUAL DIGITS OF A ‘N’ DIGIT NO.



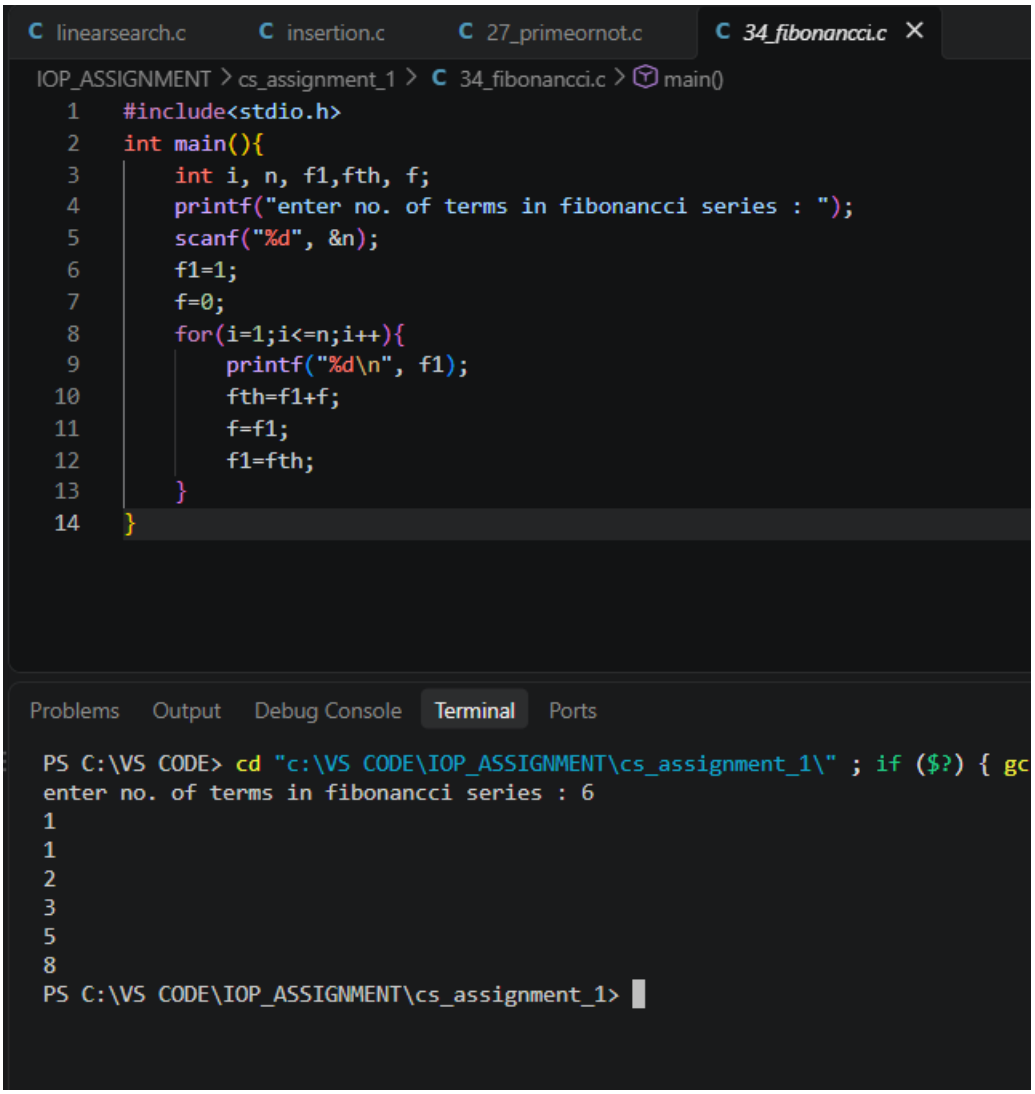
## 32. PROGRAM TO FINDOUT MAX AND SECOND MAX FROM ‘N’ NOS .



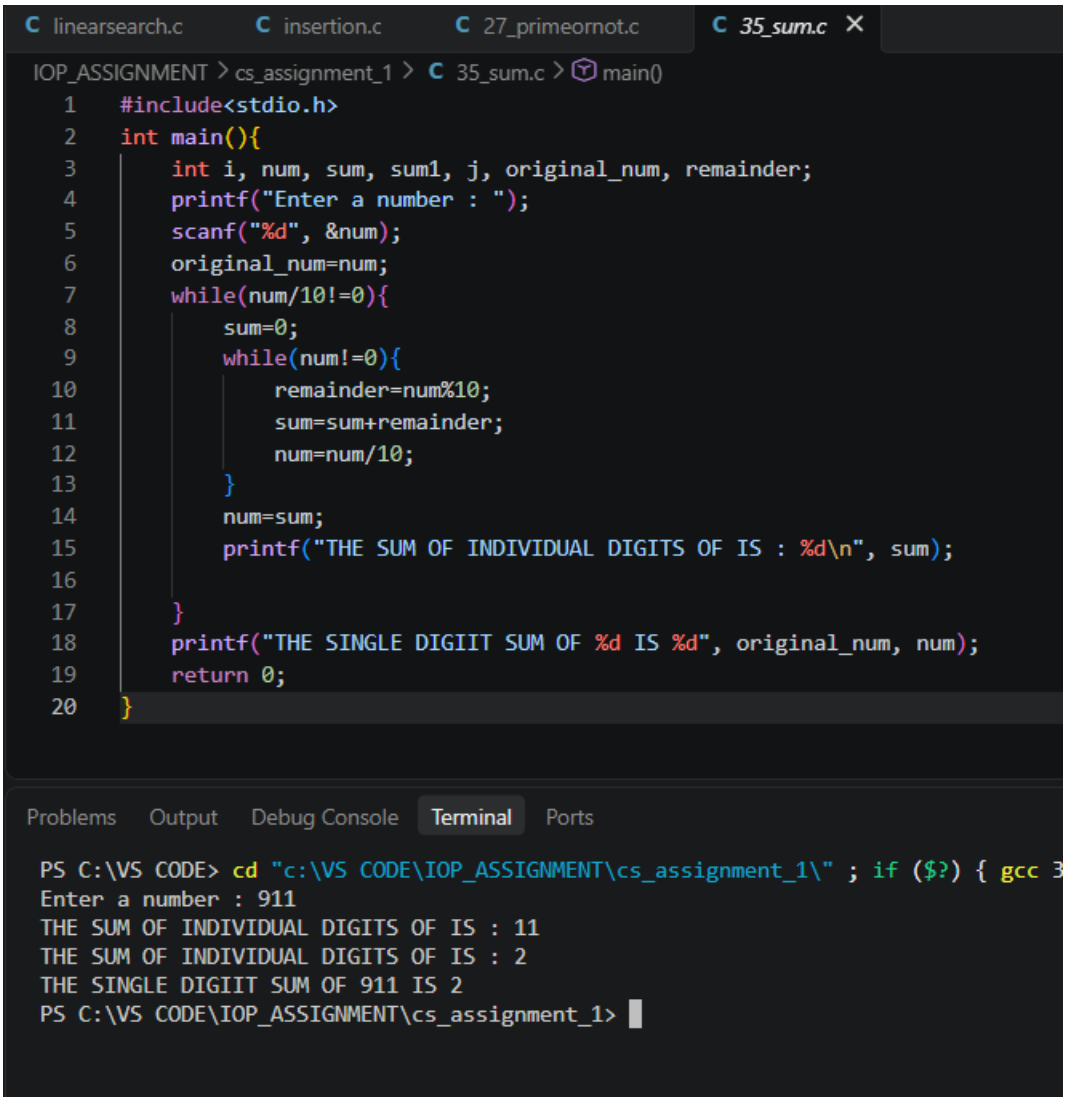
## 33. PROGRAM TO FINDOUT MAX AND MIN FROM ‘N’ NOS.



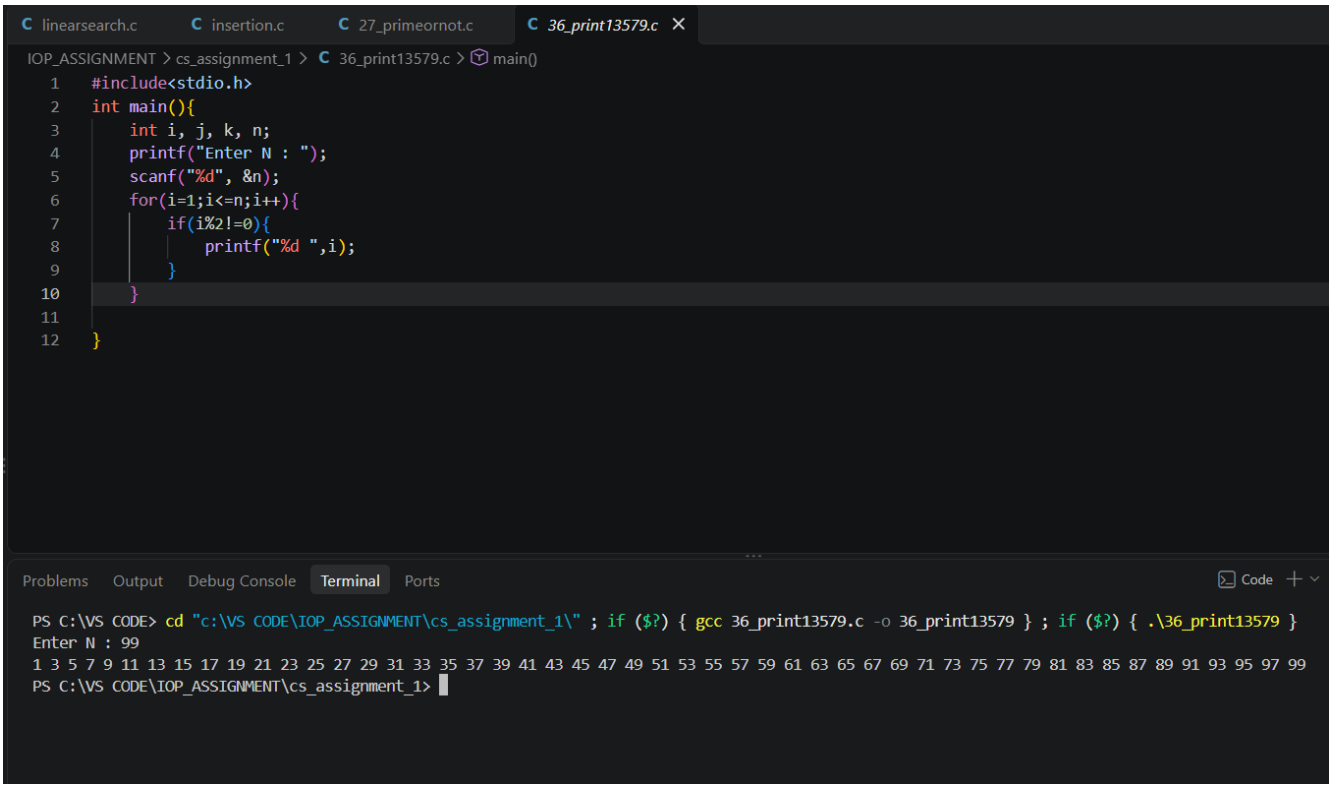
## 34. PROGRAM TO PRINT FIBONACCI SERIES. (0,1,1,2,3…)



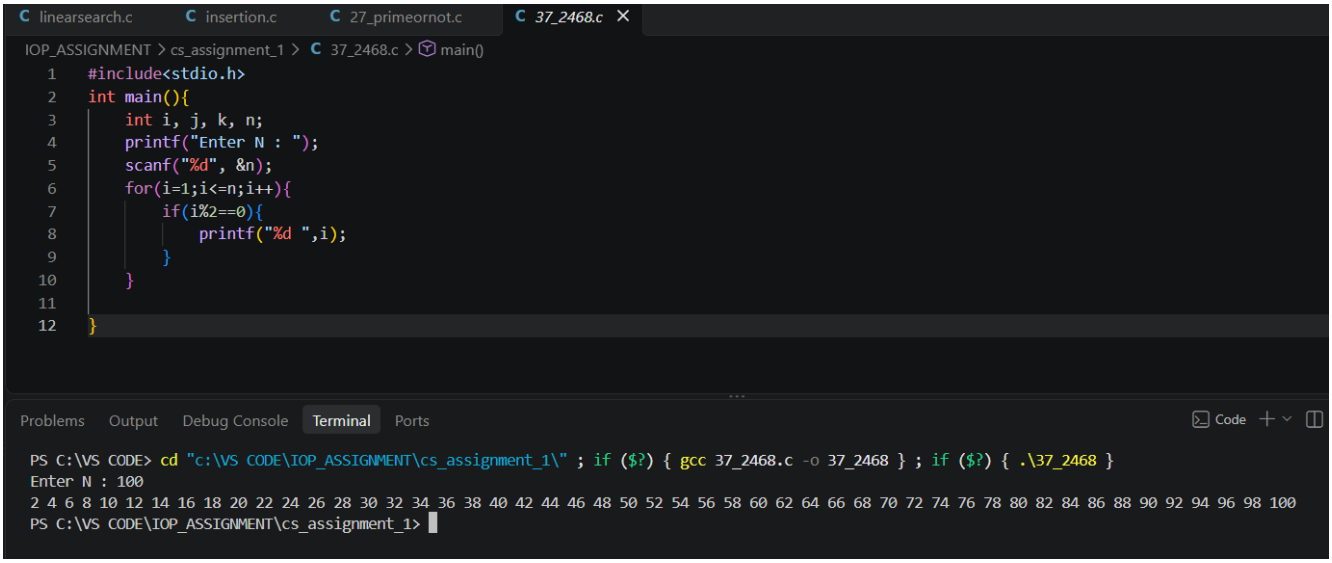
## 35. READ A NUMBER N AND PRINT A SINGLE DIGIT ANSWER SHOWING SUM OF DIGIT OF N



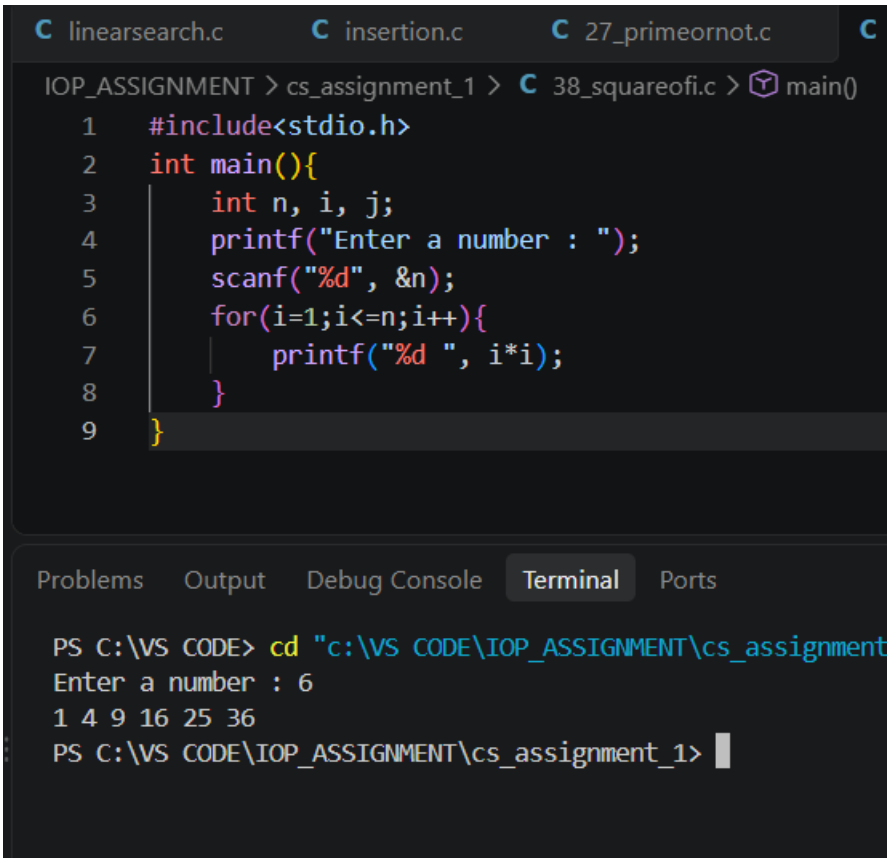
## 36. PROGRAM TO PRINT 1,3,5,7,9……N.



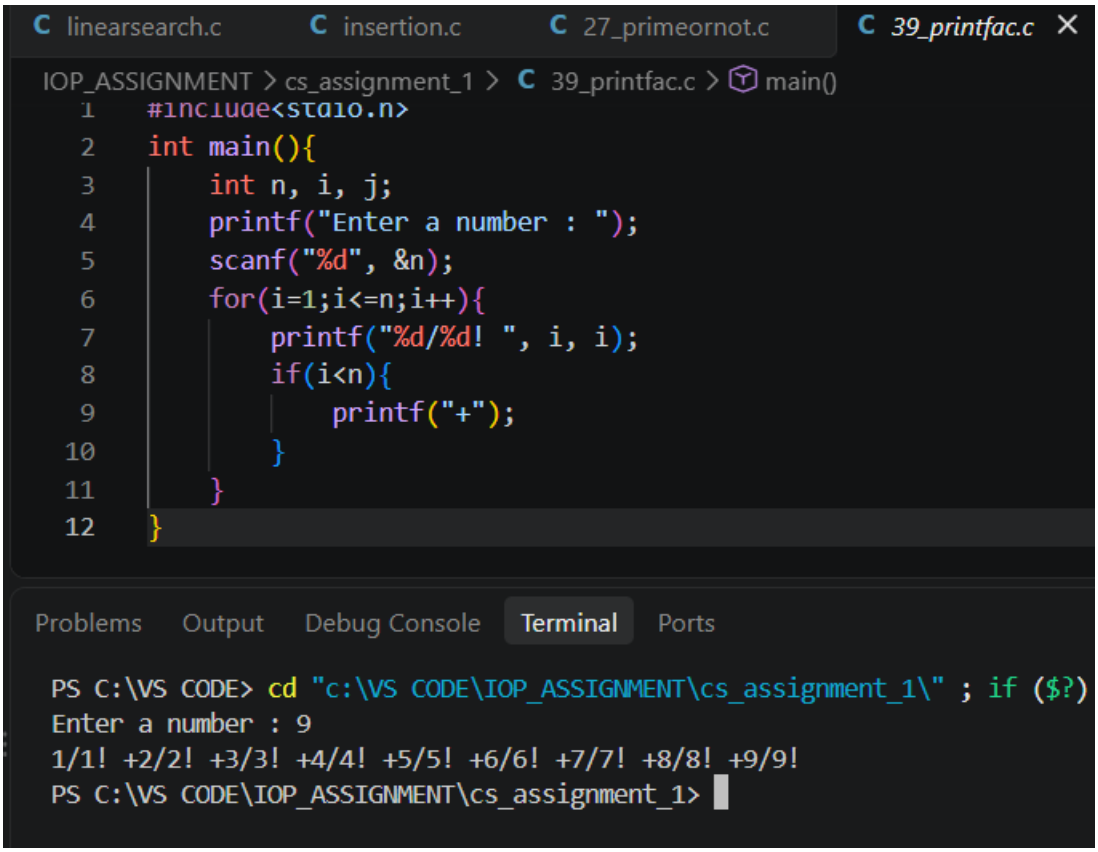

## 37. PROGRAM TO PRINT 2,4,6,8,10,12……N.



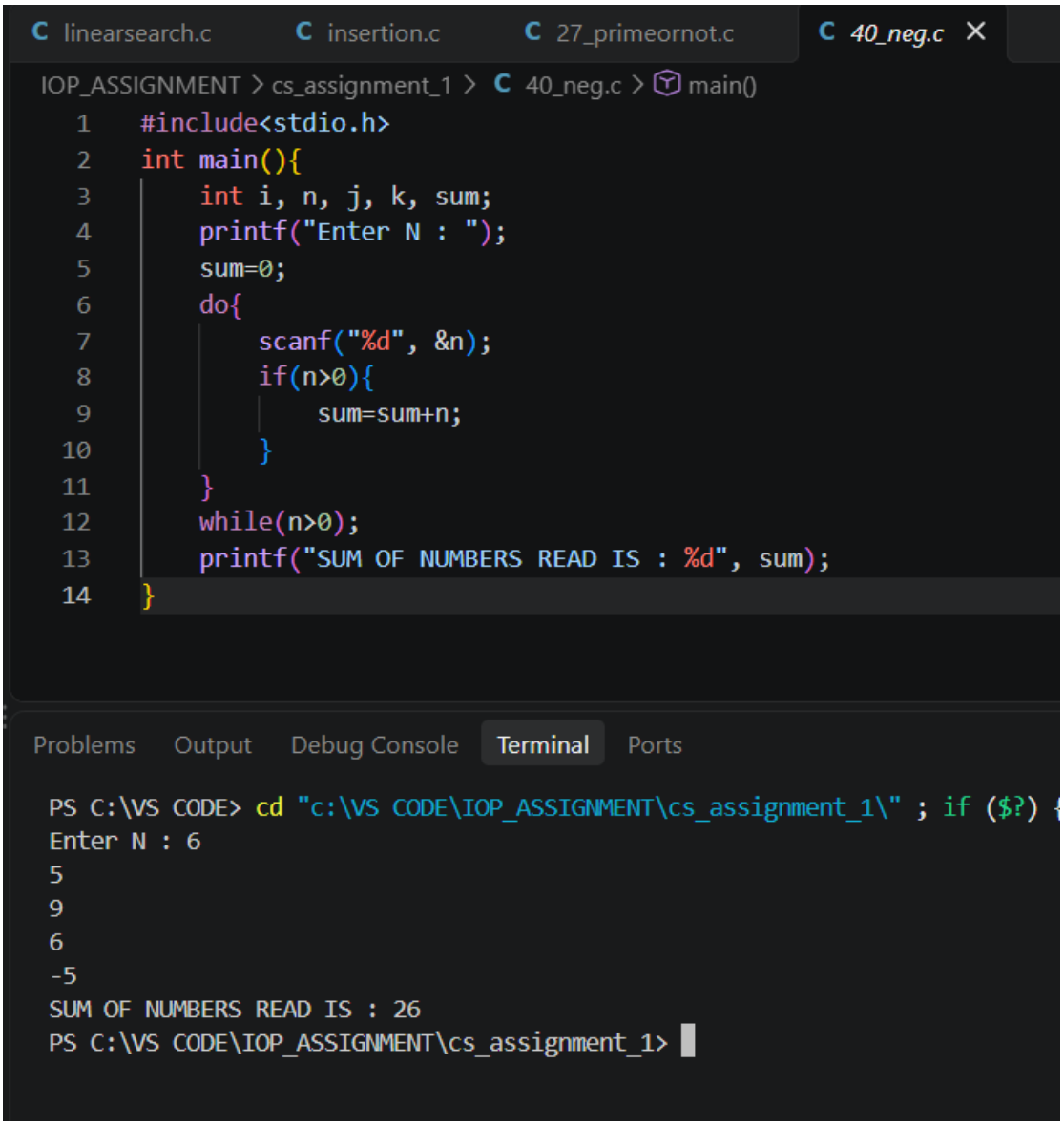
## 38. PROGRAM TO PRINT 1,4,9,16,25,……N.



## 39. PROGRAM TO PRINT 1/1!+2/2!+3/3!+4/4!,……N.



## 40. READ NUMBERS TILL A NEGATIVE NUMBER IS ENTERED AND CALCULATE SUM OF A LIST OF NUMBERS READ.



## 41. READ A NUMBER N AND PRINT FACTOR OF N.(EX 28= 1,2,4,7,14,28)

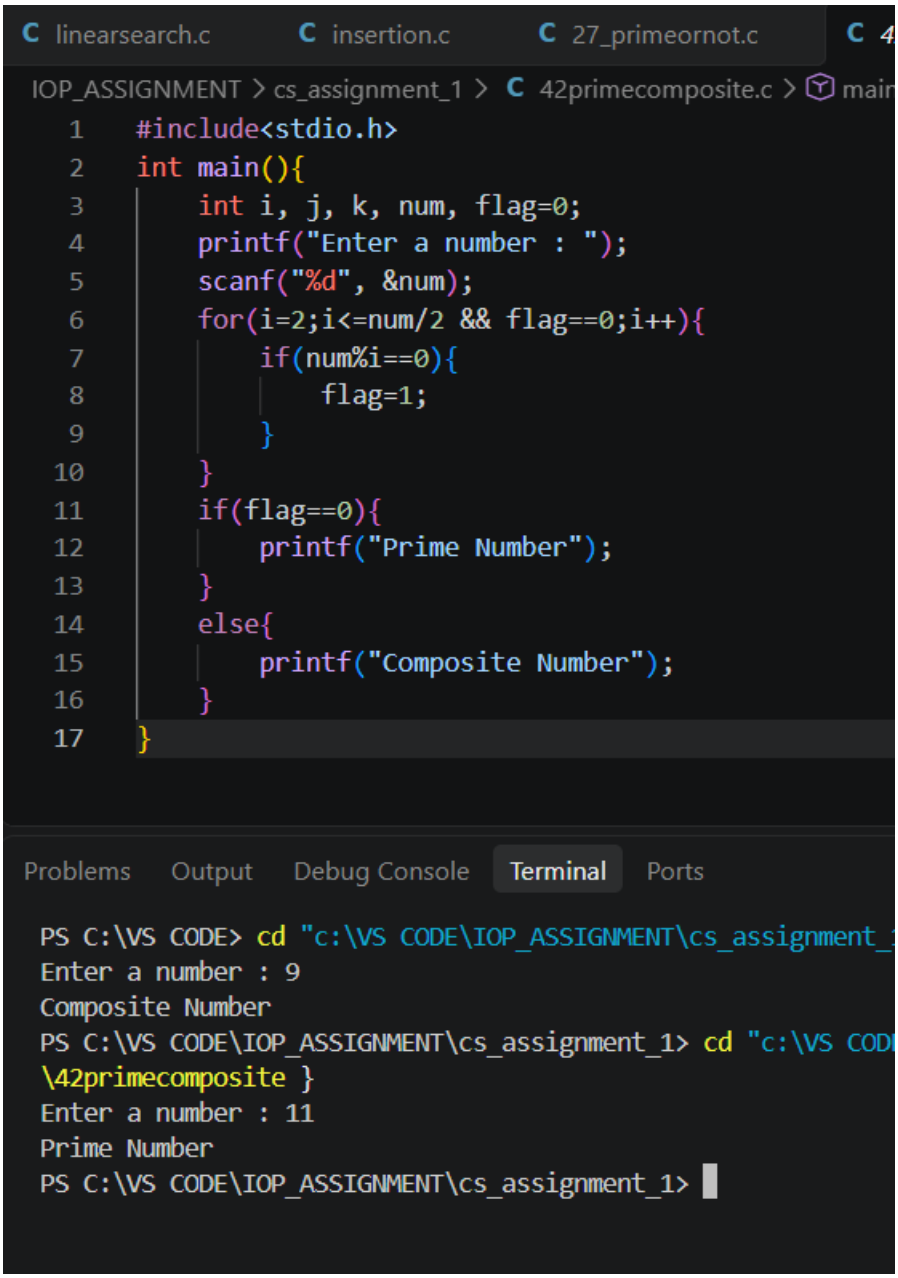


```c
int main(){
    int i, j, k, num;
    char ch;
    do{
        printf("Enter a Numbr : ");
        scanf("%d", &num);
        printf("Factors of %d are : ", num);
        for(i=1; i<=num; i++){
            if(num%i==0){
                printf("%d\t", i);
            }

        }
        printf("\n");
        printf("Do you want to continue : ");
        scanf(" %c", &ch);
    }while(ch=='y');
}
```

```
PS C:\VS CODE> cd "c:\VS CODE\IOP_ASSIGNMENT\cs_assignment_1\" ; if ($?) { gcc
Enter a Numbr : 28
Factors of 28 are : 1   2       4       7       14      28
Do you want to continue : y
Enter a Numbr : 56
Factors of 56 are : 1   2       4       7       8       14      28      56
Do you want to continue : n
PS C:\VS CODE\IOP_ASSIGNMENT\cs_assignment_1>
```

## 42. READ A NUMBER N AND PRINT IF N IS PRIME OR COMPOSITE.



43. WRITE A C PROGRAM TO CALCULATE THE COMMISSION FOR A SALES REPRESENTATIVE AS PER THE SALES AMOUNT GIVEN BELOW.

**If sales<=Rs. 500, commission is 5%**
**If sales> but <=2000, commission is Rs. 35 plus 10% above Rs. 500**
**If sales>2000 but <=5000, commission is Rs. 185 plus 12% above Rs. 2000**
**If sales>5000, commission is 12.5%**

linearsearch.c | insertion.c | 27_primeornot.c | 43s

IOP_ASSIGNMENT > cs_assignment_1 > 43salescomission.c > main()

```c
#include<stdio.h>
int main(){
    int i, j, sales, comission, pay;
    printf("Enter Sales");
    scanf("%d", &sales);
    if(sales<=500){
        comission=0.05;
        pay=sales*comission;
    }
    if(sales>500 && sales<=2000){
        comission=0.1;
        pay=(sales-500)*comission+35;
    }
    if(sales>2000 && sales<=5000){
        comission=0.12;
        pay=(sales-2000)*comission+185;
    }
    if(sales>5000){
        comission=0.125;
        pay=(sales-5000)*comission;
    }
    printf("PAY IS %d", pay);
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\VS CODE> cd "c:\VS CODE\IOP_ASSIGNMENT\cs_assignment_1\
Enter Sales2500
PAY IS 185
PS C:\VS CODE\IOP_ASSIGNMENT\cs_assignment_1>
```

44. WRITE A PROGRAM IN C FOR THE FOLLOWING.
AN ELECTRIC DISTRIBUTION COMPANY CHARGES ITS DOMESTIC CONSUMERS AS FOLLOWS.

| Consumption in unit | Rate for Charge |
|---|---|
| 0-200 | Re 0.50 per unit |
| 210-400 | Rs. 100 plus Re 0.65 per unit excess of 200 |
| 401-600 | Rs. 230 plus Re 0.80 per unit excess of 400 |
| Above 600 | Rs. 425 plus Rs. 125 per unit excess of 600 |

Print the amount to be paid by the consumer.

